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A Review of Progress in the Chemotherapy and Control of Filariasis Since 1955 *

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There has been little change since 1955 in the laboratory techniques for seeking new antifilarial compounds, although one valuable addition to laboratory study has been the experimental infection of cats with Brugia malayi.

The chief drug for the treatment and control of filariasis—whether caused by Wuchereria bancrofti *or by* B. malayi*—continues to be diethylcarbamazine, and the author reviews the reports recently published on its use. In India and China large-scale campaigns involving the use of this drug have been undertaken ; and in Tahiti filariasis has been suppressed and almost eliminated. Campaigns on a smaller scale and pilot projects considered in this survey include those conducted in Pacific islands, Malaya, Ceylon, Brazil, Surinam and East and West Africa.*

It is generally agreed that the administration of diethylcarbamazine produces a great diminution in the microfilarial counts of those taking it, and in many persons both microfilariae and adult worms are eradicated. The difficulties which arise are due to toxic effects which occur in some recipients and which may adversely affect the acceptability of treatment.

INTRODUCTION

Experimental chemotherapy

Little change in the experimental chemotherapy of filariasis has taken place since 1955. For chemotherapeutic screening, the chief organism is still *Litomosoides carinii* growing in cotton rats (*Sigmodon hispidus*). Rohde (1959) has claimed that this worm can also be maintained in white rats, which are cheaper and more widely available than cotton rats. But the reviewer and others have always found in the past that infections of *L. carinii* in white rats were light, short-lived, and generally unsatisfactory. Other experimental filarial infections have been described, including the following:

Dipetalonema witei (*D. vite*, *D. blanci*) in the jird, *Meriones libycus*; but this is not much more convenient than *Litomosoides* and it does not respond to diethylcarbamazine, showing that its chemotherapeutic reactions differ from those of human filariae.

Brugia malayi and *B. pahangi* in cats (Edeson & Laing, 1959). *B. malayi* has the great advantage of being a human filaria, but its maintenance in cats seems somewhat difficult and the cat is too large an animal for screening purposes. Nevertheless this infection is a valuable addition to the laboratory.

Dirofilaria uniformis in cotton tail rabbits, transmitted by *Anopheles quadrimaculatus*, has also been described by Moon et al. (1960) as a potential filaria for experimental laboratory infections.

Filaricidal compounds

Diethylcarbamazine. The chief compound for the therapy of bancroftian and malayan filariasis continues to be diethylcarbamazine. World-wide experience, during which this compound has been been administered to several million persons, has confirmed its safety; no fatal case of idiosyncrasy to it has ever been encountered. It has long been believed on indirect evidence that it can kill or permanently sterilize the adult worms of *W. bancrofti* and *B. malayi*. Direct evidence that diethylcarbamazine actually kills the adult worms of *B. malayi* has been provided by Edeson & Laing (1959) using cats experimentally infected with the human

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parasite. A valuable review on diethylcarbamazine has been written by Singh & Raghavan (1957).

Arsenical compounds. It was discovered in 1947 by Otto & Maren that arsenical compounds are actively filaricidal both for adult worms and for microfilariae, in particular the compound known as arsenamide, Thiacetarsamide, Thioarsenol, or Caparsolate sodium, which is a soluble thioglycollate derivative of benzamide *p*-arsenoxide. This compound, however, was toxic to the liver in patients who have an idiosyncrasy and it has not come into general use. More recently attention has been drawn to the potential filaricidal value of another compound called Mel W (pentylthiarsaphenylmelamine), which is a water-soluble derivative of the compound Mel B (melarsoprol, Arsobal) which has proved valuable for the treatment of human trypanosomiasis (sleeping sickness). Five cases of filariasis were treated with Mel W by Friedheim & De Jongh (1959). Doses of 0.1-0.2 g were given intramuscularly for 3-4 consecutive days, and the microfilarial count diminished rapidly in the next 2-3 months. Since then, extensive trials of this compound against filariasis have been carried out by Dr Friedheim in East Africa (private communication), the drug being given in 1-3 intramuscular injections. This compound apparently deserves careful consideration as a possible alternative to diethylcarbamazine in the mass control of filariasis.

Bis-isoquinolinium compounds. Antifilarial activity against *Litomosoides* was reported in compounds of this chemical series by Hawking & Terry (1959). On further investigation, however, it has been found that the compounds are too irritant for injection into man either intramuscularly or intravenously. Accordingly they are not likely to be of clinical value as filaricides.

CONTROL OF FILARIASIS BY DIETHYLCARBAMAZINE

Since 1955, interest in the control of filariasis by diethylcarbamazine has gradually become more widespread. This is illustrated by the number of abstracts on the subject appearing in the *Tropical Diseases Bulletin* : none in 1955, 4 in 1956, 2 in 1957, 1 in 1958, 8 in 1959, and 8 in 1960.

A review is accordingly given of the reports on this subject which have become available from the different parts of the world; these are summarized in Table 1. (The practice of quoting doses as the citrate salt (which contains only 51% of the active base) has now become almost universal; unless otherwise indicated all doses mentioned below refer to the salt.)

Pacific Ocean

Tahiti. The work of the Franco-American team in controlling (and practically eradicating) filariasis in Tahiti (population 30 000) is well known. It has been described by Kessell (1957) and a more recent report on the clinical aspects has been given by March et al. (1960). Only a brief outline need be included here. The chief dose-schedule was 6 mg/kg once monthly for 12 months; this was given to all the people in a district and repeated for several years. The results are illustrated by comparing the figures for 15 specially studied districts for 1949 and 1959 respectively (the male population was 3390 and 4262 in these two years):

	1949	*1959*
Microfilariae-positive	38%	6.5%
Lymphangitis	36%	4%
Hydrocele	17%	8%

During this work mass treatment was well received and it gradually became more acceptable as people who previously suffered from lymphangitis reported that this ailment was much less frequent after treatment. In the last report (March et al., 1960) it is concluded that " diethylcarbamazine with vector control is highly satisfactory as a rapid and practical way to control and eventually to eradicate the disease ".

Niue Island (Simpson, 1957). Starting from January 1956, 50 mg diethylcarbamazine were given monthly to all persons over 6 years old. People who responded with allergic reactions (i.e., microfilaria positives) were given extra courses of 50 mg thrice daily for 14 days; this cleared the blood of microfilariae. After 10 months, the microfilaria positivity rate fell from 22% (original level) to 2.9%. (Some mosquito control measures were also taken.)

Fiji. Before 1956 various small trials were carried out in Fiji, including one on Makogai Island (unpublished) with encouraging results (Nelson & Cruikshank, 1955). Studies on mosquito control were made by Symes (1960). More recently, Burnett & Mataika (1961) have described a larger trial carried out in 1958-59. Six doses of 400 mg diethylcarbamazine citrate each were given at weekly intervals (weight of adults 60-67 kg) followed by a second course six months later. Altogether 1226 people completed the first course (out of 1303 who started) and 911 completed the second course (out of 1296 who started); acceptance was clearly good. The results were as follows:

	Microfilaria rate	*Diminution of microfilaria count*
Before treatment	12.2%	
4 months after 1st course	6.7%	98%
4 months after 2nd course	2.7%	99.2%

The usual reactions were noted, but apparently they were not troublesome. Many people said that the drug made them feel drowsy and weak, and they preferred to take it in the evening and sleep it off (*cf.* the Chinese practice of giving a large single dose in the evening). After the second course of treatment infection of mosquitos with *W. bancrofti* was greatly reduced (and may have ceased), although infections with *Dirofilaria immitis* were unchanged.

Western New Guinea. A well-organized trial of mass treatment with diethylcarbamazine has been carried out by van Dijk (1961) at Inanwatan, where filarial disease (*W. bancrofti*) was severe. The compound (citrate) was given at 6 mg/kg once monthly for 12 doses to 1250 people. Acceptance was good and over 95% of persons took 11 or more doses. Side-effects occurred only after the first dose and they were relatively unimportant. At the end of this period the microfilaria rate was reduced by 94% and the total number of microfilariae was reduced by 98.7%.

India

In India 36 million people are exposed to *W bancrofti* infection and 4 million to *B. malayi*, and the largest filariasis control programme in the world has been carried out in that country. Numerous reports on different areas have been issued, but a few brief notes and examples may be quoted for comparison with results described from other parts of the world.

The standard treatment in India is 4 mg salt/kg (i.e., 200 mg/adult) daily for five days given, as far as possible, to all persons in a district. By December 1956, this course had already been given to 200 000 persons and by 1959 to 2 million persons. According to the valuable review by Singh & Raghavan (1957) the cost per head for drugs was Rs 2.5 and administrative costs Rs 0.75 per head, i.e., about Rs 3.25 per head. Mosquito control measures are also taken.

India is so vast that general figures are misleading' but reports on representative areas can be given as follows:

Uttar Pradesh. In Ballia town and rural district, Uttar Pradesh, 283 000 people took the drug for 1-5 days (181 000 for the complete 5 days), i.e., almost all the population.[1] Rural populations collaborated better than urban ones. About 15% of the people had some reactions, consisting mostly of pyrexia, pain all over the body and headache. Results are shown in Table 2. During 1958 similar control measures were started in five other similar districts in Uttar Pradesh.

Bombay province. Filariasis control measures in Surat district, Bombay province, are described by Patel & Paranjpey (1958). About 620 830 persons were offered the drug; 291 986 (47%) took some doses and 20% took the full course. Here again, collaboration was better in rural than in urban areas. The infection rate ranged from 10% to 16% in different parts, and in Surat town it had not been reduced by anti-mosquito measures employed from 1951 to 1956. Reactions occurred in 29% of people, and after several months it became very difficult to continue to persuade people to take the drug. The effects on the infection rate were difficult to evaluate but in two small groups of 70-100 persons who had taken the full course, the infection rate fell from 18% and 38% to 3.7% and 9%.

Kerala. The above campaigns refer to *W. bancrofti.* Trials in a *B. malayi* area were reported in an unpublished communication to the reviewer by the Filaria Control Unit which worked at Shertallai, in Kerala. During 1957 852 people in Kadakkarapally took the drug for 5 days, and 1039 took it for 1-5 days. The results of surveys are shown in Table 3. In a control area where no treatments were made, the figures for 1957 and 1958 showed no substantial change.

Ceylon

Work in Ceylon has been described by Dassanayake.[2] It is based on the finding of microfilariae carriers and their treatment by drug.

Finding of microfilariae carriers. Night blood films are taken of clinical cases and of their contacts

[1] Singh, M. V. (1958) *Filariasis in Uttar Pradesh and its control* (unpublished working document WHO SEA/Fil. Tour/6).

[2] Dassanayake, W. L. P. (1958) *Filariasis and its control in Ceylon* (unpublished working document WHO SEA/Fil. Tour/2).

TABLE

SUMMARY OF DIETHYLCARBAMAZINE TRIALS

Country	Authors	Dosage schedule
PACIFIC OCEAN		
Tahiti	March et al. (1960)	6 mg/kg monthly for 12 months to all persons
Niue Island	Simpson (1957)	50 mg monthly to all persons
Fiji	Burnett & Mataika (1961)	400 mg salt weekly for 6 weeks. Repeated after 6 months
Western New Guinea	van Dijk (1961)	6 mg/kg monthly for 12 months
INDIA		
Uttar Pradesh (Ballia)	M. V. Singh [a]	4 mg/kg daily for 5 days
Kerala (Shertallai)	Filaria Control Unit [b] (*B. malayi*)	4 mg/kg daily for 5 days
CEYLON	Dassanayake [c]	100 mg 3 times daily for 7 days to microfilariae-positives only
CHINA	Epidemic Prevention Station, Fukien (1959)	1 g (or 1 g + 0.5 g)
	Hsieh et al. (1960)	1 g
	Hsieh et al. (1960)	1 g
	Li (1959)	200 mg 3 times daily for 6-7 days
JAPAN	Sasa et al. (1959)	20-60 mg/kg in 10-13 days
MALAYA	Wharton et al. (1958)	5 mg/kg weekly × 6
		5 mg/kg monthly × 6
	Turner & Sodhy (1959)	0.5-12.0 mg/kg daily to total of 40-100 mg
AFRICA		
Pate Island (Kenya)	Heisch et al. (1959)	2.2 g per person in 6 days
Ukara Island (Tanganyika)	Jordan (1959b)	100-200 mg monthly or twice monthly for 1 year
Keneba (Gambia)	McGregor et al. (1952)	5 mg base/kg for 5 days
Keneba (Gambia)	McGregor & Gilles (1960)	2.5 mg base/kg for 5 days
SOUTH AMERICA		
Surinam	van der Kuyp (1955)	2.5 mg base/kg for 5 days
Brazil	Neves et al. (1955)	6 mg/kg daily for 7 days
	Rachou & Scaff (1958)	6 mg/kg once
		12 mg/kg once

[a] Unpublished working document WHO SEA/Fil. Tour/6. [b] Unpublished communication (1958) to the reviewer.

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IN DIFFERENT PARTS OF THE WORLD

Number treated	Microfilarial rate (%)		Reduction of microfilarial density (%)	Toxic effects	Acceptability	Remarks
	Before	After				
3 390 males	38	6.5		Slight	Good	Part of mass campaign for whole island
2 800	22.1	2.9		Apparently none	Apparently good	
1 226	12.2	2.7	99.2	Not serious	Good	
1 250	22	1.3	98.7	Unimportant	Good	
283 000 (1-5 days) 181 000 (full 5 days)	9-12	3.6-7	80-91	Pyrexia, pain all over body, headache in 15% of persons	Diminished gradually	
852	17.2	5.2	87			
?	8.1 (1948)	2 (1957)		Not reported	Apparently good	Only microfilariae-positives treated
144 microfilariae-positives	100	8		Marked	Accepted	Peculiar to China?
1 052 *W. bancrofti*	100	40		Marked	Accepted	
186 *B. malayi*	100	17		Marked		
234	100	20-25		Less marked		
179 microfilariae-positives	100	26-77	88-98	Slight with *W. bancrofti*; greater with *B. malayi*	Usually fair	
78	49	14	96.5	Moderate after first dose only	Fairly good	*B. malayi*, semi-periodic
119	35	7	98.5			
156	27	0	100	Severe with doses over 7 mg	Much persuasion needed	*B. malayi*, periodic
1 200	32	16	90	Negligible?	Good	
235 microfilariae-positives	100	20	89-95	Apparently negligible		
181 microfilariae-positives	100	26	94	Moderate	Only with persuasion	
54 + 36 microfilariae-positives	100	90 59	99-90	Negligible	Good	
7 113 microfilariae-positives	100	20				Few details
183 microfilariae-positives	100	27	97			
40 microfilariae-positives	100	62	91			
40 microfilariae-positives	100	49	87			

[c] Unpublished working document WHO SEA/Fil. Tour/2.

TABLE 2

RESULTS OF FILARIASIS CONTROL BY DIETHYLCARBAMAZINE IN BALLIA, UTTAR PRADESH, INDIA

Area	Population	Before therapy			After therapy			Reduction of product [a] (%)
		Infection rate (%)	Average infestation	Product [a]	Infection rate (%)	Average infestation	Product [a]	
Urban	30 000	12.05	28.9	348	7.1	9.8	70	80
Rural	254 000	8.9	11	99	3.6	3	,11	91

[a] Infection rate × infestation.

and systematic films are also taken from the whole population of some areas. Cases detected are given 100 mg diethylcarbamazine three times a day for seven days. The blood is examined again 10 days after courses (75% *W. bancrofti* and 100% *B. malayi* cases are negative). Those still positive are treated again once or twice. Apparently all this is on a fairly small scale.

Results. It is claimed that since 1937 rural filariasis due to *B. malayi* has been totally conquered. Since 1948 bancroftial areas have been controlled and microfilarial rates have been reduced from 8.1% (1948) to nearly 2% (1957). A list of 10 *W. bancrofti* areas and seven *B. malayi* areas is given with yearly microfilarial rates but no indication of the total populations involved.

Since the initial microfilarial rates ranged from 4% to 12%, it seems that infection was low compared with many other parts of the world.

China

It has always been known that filariasis is common in many parts of China, and, in view of the vast populations involved and of conditions in the country, it must be very important. According to a recent estimate there are probably about 20-30 million people with microfilariae in the blood and a total of 40 million with some evidence of filarial infection or disease. Consequently there seems to be more filariasis in China than in all the rest of the world, except India, put together.

TABLE 3

RESULTS OF FILARIASIS CONTROL BY DIETHYLCARBAMAZINE IN KERALA, INDIA

	Drug for 5 days (852 people)		Drug for 1-5 days (1 039 people)		No drug (324 people)	
	1957	1958	1957	1958	1957	1958
Infection rate (%)	17.2	5.2	17.1	6.8	18.5	16.7
Average No. of microfilariae per 20 mm³	19.2	8.0	19.3	10.6	21.3	28.1

Vigorous steps have been taken to deal with filarial and other parasitic diseases. Many provincial antifilariasis stations have been set up, and many teams of workers have been organized in the larger medical centres. A general review is given by Li Huei-Han (1959), which appears to summarize the filariasis section of a Conference on Parasitic Diseases, held in Shanghai in 1958. This deals with all aspects of the epidemiology and control of filariasis, but only the parts dealing with treatment and drug-control will be considered here. Statistics and conclusions are often presented in ways which are different from Western practice and care must be taken to avoid misinterpretation; all the same, the vast extent of the problem and the size of the efforts to cope with it compel respect, and this work deserves most careful consideration in any world-wide review of filariasis.

Filariasis surveys and case-finding. Apparently surveys are made by taking approximately 60 mm³ of blood at night (about 9 p.m.) and examining it after dehaemoglobinization. Presumably most of the slides are stained by standard methods, but a *P'inlan* rapid stain (prepared from methyl blue R) is recommended and described by Ch'en (1959). This takes only 1-3 minutes to perform. It is also stated by the same author that experienced technicians can examine blood films immediately after dehaemoglobinization and that only those films which show microfilariae need be stained. It would seem to the reviewer that this practice might miss many cases of light infections in which only 1-2 microfilariae were present in the whole slide. Complete figures for the numbers examined are not given, but reference is made by Hsieh et al. (1960) to 122 252 people examined in a single district of Shanghai, 1.33% being found positive for *W. bancrofti*.

In rural districts where infection is due to *B. malayi*, it is suggested that the allergic response to diethylcarbamazine may be used for diagnosis. In northern Anhwei, 25-100 mg diethylcarbamazine were given to 1009 villagers at the same time as blood films were taken. Out of 209 microfilariae-positive cases, 78% showed fever, headache, dizziness or anorexia; out of 800 cases not showing microfilariae at examination, only 6.1% showed symptoms and some of these may have been latent cases.

Treatment with diethylcarbamazine. Apparently this is given only to microfilaria carriers and there is no record of treating a whole population. The usual experimentation with small groups has been carried out and two main schedules have been tried: (*a*) small doses spread over 5-7 days, and (*b*) 1-4 larger doses given in less than 30 hours (this is the procedure which is recommended).

(*a*) Many pilot experiments with small doses administered over 5-7 days are described. (Unless *B. malayi* is stated, all cases were due to *W. bancrofti*.) Thus 200 mg three times daily for 7-10 days were given to cases in Nanking Army Hospital in 1955 (Li, 1959); at the end of treatment 87% were cleared of microfilariae and 11-17 months later 93% were negative for microfilariae.

In Shantung 153 cases were treated with the same dose; at the end of treatment 85% were negative, and after three months 70% were negative. In 1958 in Tsouhsien, Shantung, 234 patients were given courses of 200 mg thrice daily or 300 mg twice daily for seven days. This cleared 75%-80% of the patients and the average number of microfilariae after treatment was reduced by 65%-79%. Courses of 0.3 g diethylcarbamazine citrate thrice daily for five or three days were given by Li Lei-Shih et al. (1959) to 153 cases of bancroftian and malayan filariasis. The clearance rates (examined six months later) were 72%-89%, which is said to be as good as after seven days of courses, and the microfilarial count (in those which remained positive) was reduced by 94%. These workers noted that microfilariae in hydroceles or lymphatic channels were not destroyed by the drug; this was also the experience of the reviewer in East Africa (Hawking, 1950).

(*b*) As to the short intensive course, according to Li (1959) doses of 400 mg four times in one day or 500 mg twice daily for two days were effective in killing adult worms (as judged by the development of lymphatic nodules) in 80% and 60% respectively in the two groups of malayan cases so treated; but the therapeutic effect on the microfilariae was not as complete as with the longer course quoted above.

In a second series, 372 cases were treated with 1.5 g in one day or 2.0 g in two days; in bancroftian cases the clearance rate was 67% immediately after treatment and 76% three months later; in malayan cases the clearance rates were 84% and 100% respectively.

The Shantung Provincial Institute for the Control of Filariasis organized treatment for 264 schoolchildren with 0.5 g twice daily for two days; immediately after treatment 69%-85% were negative, and six months after treatment 50%-78% were negative and the average number of microfilariae in positive cases was reduced from 13-47 to 3-6.

Table 4 shows results obtained in Tsouhsien, Shantung, in 1958. After three months, the clearance rates in Tsouhsien were 26%-40%. The results were compared with the smaller doses given for seven days (see above), and it is concluded that single dose treatment is not so effective as a longer course; also 22%-52% of the patients vomited.

However, the short-course treatment is regarded with favour elsewhere, especially as it is more convenient and it can be given as soon as the cases are diagnosed. In an article from the Epidemic Prevention Station, Fukien (1959), it is stated that this method is extensively used in Fukien province, and mention is made of 15 914 cases treated by a single dose of 1 g at night in one administrative region; microfilariae disappeared in 92% of cases and in the

TABLE 4
RESULTS OF SHORT, INTENSIVE COURSE OF TREATMENT WITH DIETHYLCARBAMAZINE IN TSOUHSIEN, SHANTUNG, CHINA [a]

Rural patients	Dosage	Percentage becoming microfilariae-negative	Fall of microfilarial count (%)
112	1.0 g single dose	33	90
133	1.5 g total	40	87
197	2.0 g in 30 hours	34	65

[a] Based on data in Li (1959).

rest the count was reduced by over 97%. This treatment is said to be easily accepted by the public. Drug reactions at night were less noticeable than during the day. This article concludes: " Amazing results have been obtained in less than five months. In various named places filariasis has been essentially wiped out and only the question of reactions to treatment requires further study. Although acupuncture and moxibustion are of help in suppressing the reactions this method cannot be used widely owing to the scarcity of proficient practitioners. Further studies on suppression of reactions are being actively carried out."

Other experience is described by Hsieh et al. (1960) working in districts of Shanghai where the prevalence was 1%-4%. All microfilariae-positive cases were treated, most of them in improvised wards set up in the villages. A dose of 1 g of diethylcarbamazine was given at night. The blood was re-examined 3-7 days later with the following results: of 1052 cases of *W. bancrofti*, 60% became negative and the microfilaria count fell 80%; of 183 cases of *B. malayi*, 83% became negative and the microfilaria count fell 90%.

The reactions consisted of fever, headache, boneache, general malaise, nausea and vomiting—less frequently of lymphadenitis, abdominal pain, hoarseness of voice, rash and itching:

Infection	*Fever*	*Nausea & vomiting*	*Dizziness & headache*
B. malayi	86%	39%	54%
W. bancrofti	28%	22%	58%

Most patients with *B. malayi* showed fever a few hours after taking diethylcarbamazine (*cf.* onchocerciasis); it lasted 1-5 days. With *W. bancrofti*, fever occurred on the second day and was shorter and milder. With *B. malayi*, there were often inflammatory indurations in the inguinal region or medial aspect of the thigh, and with *W. bancrofti* painless nodules in the scrotum appeared after treatment in 90% of patients; both these were presumably reactions around worms killed by the drug. (The Chinese workers have also tried the arsenical compound, carbarsone, 0.5 g by mouth, twice daily for 10 days; it was toxic and not very effective.)

In discussing all these results Hsieh et al. conclude:

" Single dose diethylcarbamazine treatment of filariasis has great practical value as well as a sound theoretical basis. Its only drawback in comparison with the conventional method of repeated smaller doses is the occurrence of gastrointestinal irritation symptoms such as nausea and vomiting. These however are not serious and can be tolerated by the patient . . . it had better effects for *W. malayi* than *W. bancrofti* . . . The method is simple, rapid, easy to be carried out and is convenient for large scale treatment in the field, especially fit for low endemic areas."

General comment. It appears to the reviewer that the percentages of *W. bancrofti* carriers reported to be made negative after a single dose of diethylcarbamazine are often appreciably higher than those found in other countries for five-day courses (which are agreed to be therapeutically more effective than single doses). Perhaps conditions in China for mass-acceptance of large nauseating doses of drug are, for a number of reasons, more favourable than elsewhere. It is clear that filariasis is being studied and combated on a vast scale, and that the experience and methods of Chinese workers deserve serious consideration.

Japan

In Japan several different teams have studied filariasis, especially in outlying islands (Sasa et al., 1959). So far only pilot experiments with diethylcarbamazine have been reported, although apparently it is intended that these shall form the basis of larger campaigns.

In eight villages with populations ranging from 73 to 1134 the microfilarial rates were 0.5%-23%. All the people were treated according to various schedules ranging from 2 mg/kg daily for 10 days (total 20 mg) to similar schemes totalling 60 mg/kg in 13 days. In these villages 23%-64% of the microfilarial carriers became negative 120 days later, and the microfilarial counts in the others fell by 88%-98%. In other villages 2 mg/kg were given every week or every month for 10 doses; this appeared to be more effective, but the numbers treated were small. The recommended treatment is either 2 mg/kg daily for five days or 2 mg/kg every 10 days for 10 times.

Malaya

The important work carried out on filariasis by Wilson, Edeson, Wharton, Reid and their colleagues is well known and has been widely reported. Accordingly the part of their work which concerns diethylcarbamazine will be summarized here more briefly than its importance merits.

B. malayi, *semi-periodic form, Pahang region.* Various dose schedules were tried on small groups of patients and the dose recommended for mass

TABLE 5
EFFECT OF DIETHYLCARBAMAZINE ON SEMI-PERIODIC FORM OF *B. MALAYI* INFECTION IN MALAYA[a]

Village	Dose schedule	No. of persons	Time	Microfilarial rate (%)	Diminution in number of microfilariae in all persons (%)
Sawah	0.5 mg/kg, 1.0 mg/kg, 5.0 mg/kg × 6 weekly	78	Start	49	
			After 1 year	14	96.5
Tanah Puteh	5 mg/kg × 6 monthly	119	Start	35	
			After 1 year	7	98.5

[a] Based on data in Wharton et al. (1958).

treatment was 5 mg salt/kg weekly or monthly for six doses. This dosage made all persons microfilariae-negative six months after the last dose. If the first dose was a small one, reactions were less severe (Edeson & Wharton, 1958). Trials were then carried out in two villages in the Pahang region (Wharton et al., 1958) with the results shown in Table 5. In a third village, sprayed with dieldrin, the figures for filarial infection showed no change in one year.

Toxic reactions were only moderate; they occurred mostly after the first dose. The weekly dose was somewhat easier to give than the monthly one (Wharton et al., 1958).

B. malayi, *periodic form, Penang and Kedah areas.* Various preliminary trials were carried out by Turner (1959); and then Turner & Sodhy (1959) treated 156 people in the Penang area (*B. malayi*, periodic form). The dosage was 0.5-2 mg salt/kg, increasing daily up to a maximum single dose of 12 mg/kg and a total dose of 40-100 mg/kg; but there were many irregularities owing to reactions and defaulting.

The microfilarial rate before treatment was 27% and 41 weeks after treatment no microfilariae were found (one person had been found microfilariae-positive four weeks after the start of treatment and he had been given a further treatment totalling 69 mg/kg). Febrile reactions occurred in all microfilariae-positive persons and in 19% of others. People did not tolerate doses greater than 7 mg/kg at a time. Much persuasion was needed to induce people to finish the course of treatment.

In Kedah during 1956 Hassan (1959) conducted a mass campaign involving 3529 persons, including infants. Doses of 5 mg salt/kg were given weekly for six weeks. Before treatment the microfilarial rate was 26.2%, and 7-12 months after treatment was 1.5%; the total microfilarial count had been reduced by 93%. Febrile reactions occurred in all microfilariae-positive cases and in 20% of the others; also debilitating lymphangitis occurred during the third and fourth weeks in 1%-2% of persons. Nevertheless, co-operation was good.

Africa

Pate Island, Kenya. A small-scale eradication campaign was carried out on Pate Island (on the sea coast) by Heisch, Nelson & Furlong (1959). All the 1200 adults were given 2.2 g per person during six days. The microfilarial rate fell from 32% before treatment to 25% two weeks later and to 16% eight months later; and the microfilarial density fell by 90%. Toxic reactions are not reported, and apparently this dosage was acceptable to the local population.

Ukara Island, Tanganyika. A pilot trial was carried out on Ukara Island (in Lake Victoria) by Jordan (1959b). Two hundred and thirty-five microfilariae-positive persons were treated by various schedules of 100 mg or 200 mg monthly, or 200 mg every second month for 1-2 years. At the end of the first year 80% were cleared of microfilariae and the reduction of the microfilarial count was 89%-95%. It was considered that 100 mg monthly is too low a dose and that microfilariae-positive persons should receive 400-600 mg monthly.

Gambia. Pilot experiments on the control and eradication of *W. bancrofti* in a village (Keneba) were started by McGregor, Hawking & Smith (1952) in 1951 and they have been continued by McGregor &

Gilles (1956, 1960). Briefly, the population of 603 persons was examined and 181 persons who were microfilariae-positive were given diethylcarbamazine 5 mg base/kg for five successive days; minor toxic reactions tended to make treatment unpopular. Nine months later, 65% of these carriers were microfilariae-negative, and three years and nine months later 74% were negative. The total microfilarial count of all persons was reduced (at nine months) by 94%. In 1955, 54 people who were still microfilariae-positive in Keneba and 36 similar people in the adjacent village of Jali were treated with 2.5 mg base/kg on five successive days (i.e., half the previous dose). This dosage was quite acceptable to the patients. Thirty months later 90% of the patients in Keneba were negative and 59% of those in Jali. The total microfilarial count was diminished by 89%-90%. It is concluded that "diethylcarbamazine in an effective dosage acceptable to a population could unaided (i.e. without antimosquito measures) do much to eradicate bancroftian filariasis in West Africa", and that mass campaigns to dose all inhabitants every 2-4 years might prove the most effective and economical way of dealing with this infection.

It is noteworthy in these studies that the figures for patients cleared of microfilariae tended to improve in the period between nine months and four years.

South America

Brazil. The control of filariasis in Brazil has been studied for many years, although judging by the published papers operations involving diethylcarbamazine have been mostly on a small scale.

In 1955 Neves, Rachou & Scaff reported the results of trials in which microfilariae carriers were given 6 mg/kg daily for various periods, with the results shown in Table 6. The results obtained after seven days were almost as good as those after 21 days.

Rachou (1957) reports on the filariasis control programme carried out in Brazil by the National Department for Rural Endemic Diseases. In general it has been found that vector control (*Culex pipens fatigans*) is difficult, complex and expensive, but that diethylcarbamazine, given to microfilarial carriers in order to reduce the reservoir of infection, is a good prophylactic. Accordingly the general procedure of the Department was summarized as follows:

"1. With the present economic possibilities of the [Department], the control of bancroftiasis is being done in Brazil almost exclusively by the reduction of the number of infection sources, which is obtained with [diethylcarbamazine].

"2. The [Department] recognizes that acting simultaneously against the vector, filariasis control would be more efficient and shortened.

"3. Vector control is carried on only where the [Department] receives local financial aids and effective cooperation for improving sanitary conditions.

"4. In the solution of the vector problem, the participation of the exposed inhabitants, obtained by methods and techniques of health education, is required."

TABLE 6

REDUCTION IN MICROFILARIAL COUNT FOLLOWING TREATMENT WITH DIETHYLCARBAMAZINE (6 mg/kg) IN BRAZIL [a]

Days of treatment	No. of persons treated	Percentage negative after 12 months	Diminution of total microfilarial count after 12 months (%)
7	183	73	97
10	186	77	98
15	158	83	98.4
21	171	90.2	99.9

[a] Based on data in Neves et al. (1955).

The results of this policy are illustrated by some figures obtained in Ponta Grossa, in the municipality of Florianópolis, State of Santa Catarina (numbers involved are not given). In December 1953, microfilarial carriers and cases of elephantiasis were given diethylcarbamazine 6 mg/kg daily for seven days, and this treatment was repeated annually in December (Table 7).

Rachou & Scaff (1958) recall the results of Neves, Rachou & Scaff (1955) above and report further results obtained with schedules of 6 mg/kg on one or on three days, and of 12 mg/kg on a single day (Table 8).

The results in Table 8 are not so good as after courses lasting 7-21 days, but they are still quite valuable and the task of administration is much less. Accordingly, it is recommended that control should be simplified by omitting blood examination of most of the people and by giving mass therapy consisting of a single dose of 6 mg/kg (approximately 0.4 g per adult) to the whole population of an infected district every 6 or 12 months. This procedure is already being tested in an unnamed locality in the south of

TABLE 7

REDUCTION OF RESERVOIR OF BANCROFTIAN FILARIASIS IN *CULEX FATIGANS* IN BRAZIL BY DIETHYLCARBAMAZINE TREATMENT OF MICROFILARIAL CARRIERS [a]

Indices	Before 1953	After		
		1954	1955	1956
Index of microfilariaemia (%) [b]	14.5	11.2	7.5	2.5
Average microfilariaemia [c]	3.0	0.5	0.2	0.03
Index of infestation of *C. fatigans* (%)	10.8	1.1	0.6	0.6
Index of infectivity of *C. fatigans* (%)	0.7	0.8	0	0

[a] Based on data in Rachou (1957).
[b] Microfilarial rate?
[c] Apparently based on all slides.

Brazil. (This single-dose treatment is like the Chinese practice.)

Surinam. Treatment with diethylcarbamazine according to three different schedules was given to 7113 microfilariae-positive persons; and one year later, 80% were microfilariae-negative (van der Kuyp, 1955).

British Guiana. Between 1955 and 1961 the bloods of 223 000 persons were examined and 17 700 (7.6%) were found to contain microfilariae. Treatment was given as 0.4 g diethylcarbamazine (citrate) daily for two days; one day (Sunday) no treatment; 0.8 g for three days, and 1.2 g for two days. Doses were given once daily under supervision to most of the microfilaria carriers. Acceptance was relatively good. In 1713 cases who were re-examined one year after treatment there had been no recurrence of infection (Dr W. E. Adams—personal communication). In 1961, a Filariasis Research and Control Unit was set up under a Joint Fund Agreement between the Government of British Guiana and the Agency for International Development (AID).

TABLE 8

REDUCTION IN MICROFILARIAL COUNT FOLLOWING TREATMENT WITH DIETHYLCARBAMAZINE (6 mg/kg OR 12 mg/kg) GIVEN IN A SINGLE DAY IN BRAZIL [a]

Dose	No. of patients treated	Microfilarial rate (%)			Diminution of total microfilarial count after approx. 12 months (%)
		Before	Immediately after	12 months after	
6 mg/kg once	40	100	42	62	91.4
12 mg/kg once	40	100	37	49	87.1

[a] Based on data in Rachou & Scaff (1958).

SURVEY OF CONTROL BY DIETHYLCARBAMAZINE

In surveying the above reports a distinction should be made between pilot experiments and actual large-scale campaigns to control filariasis. The largest campaigns have been carried out in India; large campaigns are also in progress in China, but details are scanty. The most complete programme has been carried out in Tahiti, involving a moderate-sized rural population of approximately 20 000. Practical campaigns on a small scale have also been conducted in Malaya (Hassan, 1959), in Pate Island (Kenya) and in Niue Island (Pacific). Small campaigns have also been carried out or are being carried out in Surinam and British Guiana, but, again, information is scanty. Control through the systematic treatment of microfilaria carriers has been attempted in Ceylon. Pilot experiments on which mass campaigns may later be based have been carried out in Malaya, Japan, Fiji, Brazil, Tanganyika (Ukara Island) and the Gambia (Keneba).

Dosage schedules

Four main types have been employed:

1. An interrupted dose of 5-6 mg/kg monthly or weekly for 6-12 doses. This has been employed in Tahiti, Fiji, Malaya and Tanganyika. It seems to be more effective and less toxic than the other schedules, but it is more laborious to administer.

2. One dose (e.g., 4 mg/kg) daily for five (or seven) days; this is the standard schedule in India and it has also been employed in Japan, Kenya, Gambia and Brazil. This schedule is chosen particularly for its greater convenience of administration.

3. A single large dose, e.g., 1.0 g; this has been recommended mainly in China, and it is also recommended in Brazil. Although it is probably less effective and more toxic than the other schedules, it seems so much more convenient to administer that it probably deserves more consideration than it has received in the past.

4. Treatment of microfilaria carriers only has been employed in Ceylon, a dose of 100 mg per person being given three times a day for seven days.

Such restriction of the treatment to microfilariae-positive cases only must be evaluated by comparing the cost, labour and efficiency of detecting carriers, with the cost and acceptability of treating a whole population.

GENERAL CONCLUSIONS

The experience of the last eight years has shown conclusively that if diethylcarbamazine is administered to, and received by, a population, it is effective in greatly reducing the microfilarial reservoir available to infect mosquitos; and if efficient administration is repeated over several years, filarial infection can be practically eradicated. This treatment can be carried out without danger to life or health. On the other hand, there are sufficient minor toxic effects which make the treatment poorly accepted in some (but not all) areas. Experience seems to vary greatly in this respect, as is shown by the reports above. The greatest difficulty over acceptability seems to have been experienced in India. Difficulty is more likely to be encountered in large campaigns than in small pilot experiments. The greater the severity of filarial lesions, the less presumably will be the difficulty of acceptance, and conversely. It is possible that the public attitude will improve as campaigns are repeated in the same areas; some of the opposition may be due to mental inertia and to opposition to any new idea. (In some countries there has developed a popular belief that diethylcarbamazine may cause miscarriages; this is denied by Laigret & Voisin (1958), but they consider that it probably acts as an emmenagogue and that further investigations of vaginal cytology ought to be made in women who are receiving this compound.)

Toxic reactions following diethylcarbamazine

The toxic reactions caused by diethylcarbamazine are of two kinds:

1. Those which are due to the compound itself, e.g., headache, dizziness, sleepiness, nausea and vomiting, etc.

2. Those which are allergic reactions due to destruction of microfilariae and adult worms, e.g., pyrexia, fever, local inflammations around dead worms, pruritus, etc. These reactions occur particularly after the first dose and they are usually absent after later and larger doses. They would presumably be caused by any compound which was efficient in killing filariae. They are particularly marked with infections of *B. malayi*; but fortunately *B. malayi* is particularly sensitive to the drug and so the dose can be correspondingly reduced.

Attempts have been made in various ways to reduce these toxic reactions, but little success has been achieved. Some of the symptoms (especially sleepiness) can be reduced by giving the compound in the late evening, as is done in China. In Malaya, the first doses of a course have been much smaller than later doses. The administration of antihistamines, calcium salts, etc. along with diethylcarbamazine seems to have made little difference to the reactions. Fortunately, diethylcarbamazine never seems to have caused any dangerous reactions (e.g., agranulocytosis), even though it has now been administered on a very wide scale.

Animal reservoir

Judging by the published reports, the control of filariasis by means of diethylcarbamazine is gradually spreading as its value becomes more widely recognized and the difficulties of administration are gradually overcome. The possibility that such control may be invalidated by the presence of an animal reservoir of infection has been demonstrated only in the south-east of Malaya. In all other parts of the world, the possibility of such an animal reservoir can be ignored in practice at the present time.

Minimum level of microfilaraemia

Much discussion has taken place about the minimum level of microfilaraemia which will still permit infection of mosquitos. Thus it has been reported that a man with one microfilaria per 40 mm^3 of blood was infective to 2.6% of the mosquitos fed on him (Jordan, 1959a; see also Krishnaswami et al., 1959; Wharton, 1957). The question should be considered not in absolute terms (whether some transmission may still occur) but in quantitative ones (how many infective larvae are likely to be carried over). In filariasis one worm stays one worm, and the build-up of infection proceeds slowly, requiring many mosquito bites over a long time. The number of infective larvae carried over by mosquitos will be proportional (within limits) to the number available to enter mosquitos and to the number of mosquitos which can bite a second time two weeks after the first bite. Significant diminution of either of these numbers will produce a proportional diminution of the transmission of infection.

Immediate results obtained by diethylcarbamazine

In a long-lasting infection such as filariasis, suppression of transmission is not in itself sufficient to suppress the disease. In the Kodera Valley of Kenya, where transmission of the allied filarial infection by *Onchocerca* was prevented in 1946 by the extermination of the vector (*Simulium*), the disease still persists in 70% of adults over 20 years old (Nelson & Grounds, 1958). It is a great asset of diethylcarbamazine control of bancroftian and malayan filariasis that the drug not only diminishes transmission but also kills the worms and diminishes further development of the disease.

RÉSUMÉ

Depuis 1955, les techniques de recherche de nouveaux composés antifilariens au laboratoire ont peu évolué. Edeson et ses collaborateurs ont pu infester des chats par *B. malayi*, c'est-à-dire par une filaire humaine, pour des études de laboratoire. La diéthylcarbamazine continue d'être le grand médicament thérapeutique et prophylactique contre la filariose à *bancrofti* et à *malayi*; elle a déjà été administrée à plusieurs millions de sujets. Un nouveau composé arsenical, le Mel W (pentylthioarsphénylmélamine), introduit par Friedheim, mérite une étude plus approfondie en vue de son application possible aux campagnes de masse contre la filariose.

L'auteur passe en revue les rapports reçus depuis 1955 sur l'emploi de la diéthylcarbamazine dans la lutte antifilarienne. Pendant cette période, il a été fait une utilisation de plus en plus grande de ce médicament (souvent en association avec des mesures anti-moustiques). A Tahiti, la filariose a été maîtrisée et presque éliminée. En Inde, des campagnes à grande échelle ont été entreprises et la diéthylcarbamazine a été administrée à plus de 2 millions de sujets; en dépit des difficultés rencontrées pour faire accepter le médicament, ces campagnes se poursuivent. En Chine également, de vastes campagnes se déroulent, mais on est fort peu renseigné à leur sujet. Des campagnes à échelle réduite ont eu lieu en Malaisie, à Ceylan, au Kenya (île de Paté), à l'île de Niue (Pacifique), en Guyane hollandaise et en Guyane britannique. Des expériences pilotes ont été exécutées en Malaisie, aux îles Fidji, au Brésil, au Tanganyika (île d'Ukara) et en Gambie (Keneba).

Il est en général reconnu que l'administration de diéthlycarbamazine diminue considérablement le nombre de microfilaires chez les sujets traités et supprime complètement les microfilaires et les vers adultes chez beaucoup de personnes. Les difficultés rencontrées proviennent des effets toxiques (non dangereux) que le médicament provoque chez certains sujets; ils peuvent rendre le traitement moins acceptable par la population.

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